Version No.				
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MATHEMATICS HSSC-I

SECTION – A (Marks 20)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اوّل لازمی ہے۔اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔لیڈ پنسل کا استعمال ممنوع ہے۔

Answer Sheet No. _____

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پُر کریں۔ **Invigilator Sign.** _____

Fill the relevant bubble against each question according to curriculum: **Candidate Sign.** _____

	Question	A	B	C	D	A	B	C	D
1.	The factorization of $z^2 + a^2$ over complex numbers is:	$(z+a)(z-a)$	$z(z+a^2)$	$(z+ia)$ $(z-ia)$	$(z-ia)^2$	○	○	○	○
2.	If $z = x + iy$, then in polar form $z = r(cos\theta + i\, sin\theta)$, where r equals:	$x+y$	x^2+y^2	$\sqrt{x^2+y^2}$	$tan^{-1}\left(\frac{y}{x}\right)$	○	○	○	○
3.	Which of the following is **NOT** an elementary row operation?	Swapping two rows	Multiplying a row by a nonzero scalar	Adding a multiple of one row to another	Raising a row to a power	○	○	○	○
4.	The system $x+y+z=2, x+y+z=3, x+y+z=4$ is:	Dependent	Inconsistent	Consistent with infinitely many solutions	Consistent with a unique solution	○	○	○	○
5.	In an arithmetic sequence, if $a_1 = 3$ and $d = 2$, then 5th term is:	9	11	12	13	○	○	○	○
6.	Which of the following sequences is Harmonic?	$2, 4, 6, 8, \ldots$	$3, 6, 12, 24, \ldots$	$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$	$1, 3, 5, 7, \ldots$	○	○	○	○
7.	The coefficient of x^2 in $(1+2x)^3$ is:	12	18	24	30	○	○	○	○
8.	According to the principle of mathematical induction, the base step is to verify for:	$n=1$	$n=2$	$n=k$	$n=k+1$	○	○	○	○
9.	If $z_1 = r_1(cos\theta_1 + isin\theta_1)$ and $z_2 = r_2(cos\theta_2 + isin\theta_2)$, then z_1z_2 equals:	$r_1r_2e^{i(\theta_1+\theta_2)}$	$r_1r_2e^{i(\theta_1-\theta_2)}$	$\frac{r_1r_2}{e^{i(\theta_1+\theta_2)}}$	$\frac{r_1r_2}{e^{i(\theta_1-\theta_2)}}$	○	○	○	○
10.	The last digit of 7^{10} is:	1	3	7	9	○	○	○	○
11.	Remainder of $f(x) = x^3 - 2$ when divided by $(x-1)$ is:	−3	−1	1	2	○	○	○	○

	Question	A	B	C	D	A	B	C	D
12.	Which one of the following is a factor of $x^3 - 6x^2 + 11x - 6$?	$x+1$	$x-1$	$x-4$	$x+3$	○	○	○	○
13.	Which of the following is **WORNG**?	$sin^2x + cos^2x = 1$	$1 - cos^2x = sin^2x$	$sin(x+y) = sinx + siny$	$1 + cot^2x = csc^2x$	○	○	○	○
14.	The maximum value of $3 + 4\,sin\theta$ is:	3	4	7	−1	○	○	○	○
15.	The period of $y = sinx$ (in radians) is:	$\frac{\pi}{2}$	π	2π	4π	○	○	○	○
16.	$sin\frac{\pi}{3}cos\frac{\pi}{6} =$	$\frac{1}{4}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{4}$	$\frac{3}{4}$	○	○	○	○
17.	For what value of x, vectors $2\underline{i} - \underline{j} + 3\underline{k}$ and $x\underline{i} + 2\underline{j} - 2\underline{k}$ are orthogonal (mutually perpendicular)?	−2	1	2	4	○	○	○	○
18.	A constant force acts on an object is perpendicular to the displacement, work done is:	Maximum	Zero	Negative	$\lvert\vec{F}\rvert\lvert\vec{d}\rvert$	○	○	○	○
19.	If there is an option for 3 Shirts and 4 Trousers then number of outfits is:	12	10	7	1	○	○	○	○
20.	Number of permutations of 5 distinct objects taken 3 at a time is:	10	20	30	60	○	○	○	○

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MATHEMATICS HSSC–I

Time allowed: 2:35 Hours **Total Marks Sections B and C: 80**

SECTION – B (Marks 48)

Q. 2 Solve the following parts. **(12 x 4 = 48)**

(i)	Solve the quadratic equation $z^2 - 5z + 6 = 0$ by completing the square method.	04	OR	Given $tan\beta = 2$, find the values of $cos2\beta$ and $tan2\beta$	04
(ii)	Prove by using mathematical induction: $1+2+3+\ldots+ \quad n = \frac{n(n+1)}{2}$	04	OR	Verify that $cos^2\theta - sin^2\theta = cos2\theta$ for $\theta = \frac{\pi}{6}$	04
(iii)	Find the inverse of given matrix using row operations. $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	04	OR	Find the value of m if vectors $\underline{a} = \hat{\imath} + m\hat{\jmath} + 2\hat{k}$, $\underline{b} = 2\hat{\imath} - \hat{\jmath} + 3\hat{k}$, $\underline{c} = 3\hat{\imath} + 2\hat{\jmath} + \hat{k}$ are coplanar.	04
(iv)	Verify that: $cos\theta.\cos\left(\frac{\pi}{3} - \theta\right).\cos\left(\frac{\pi}{3} + \theta\right) = \frac{1}{4}cos3\theta$	04	OR	Find the value of k if the given system is consistent: $x - 4y + 3z = 0$, $2x + ky + z = 0$, $x - 2y + kz = 0$	04
(v)	Solve system of equations: $x + y = 6,\ x + z = 5, z = 2$ by using Cramer's Rule.	04	OR	Find values of n and r, if: **a.** $\binom{n}{2} = 28$ **b.** $\binom{10}{r} = 120$	04
(vi)	Find a Harmonic sequence whose 4th term is $\frac{1}{3}$ and 9th term is $\frac{1}{13}$.	04	OR	Expand $(1+x)^5$ using Binomial theorem and find the coefficient of x^2.	04
(vii)	Find maximum and minimum values of $y = 5 + 2sin\theta$	04	OR	Use factor theorem to factorize $x^3 - 3x^2 + 2x$	04
(viii)	Divide $x^3 + 6x^2 + 11x + 10$ by $x + 2$ and find quotient and remainder.	04	OR	Use Binomial Theorem to expand up to four terms and find an approximate value (correct up to 3 decimal places) of $(1.03)^3$.	04
(ix)	Find modulus and argument of complex number $3 + 4i$	04	OR	If $\underline{a} = 5\hat{\imath} - 2\hat{\jmath} + \hat{k}$ and $\underline{b} = \hat{\imath} + 2\hat{\jmath} + 2\hat{k}$, then find projection of $\underline{a}$ along $\underline{b}$ and projection of $\underline{b}$ along $\underline{a}$.	04
(x)	The 10th term of an arithmetic series is 25 and the first term is 7: **a.** Find the common difference **b.** Find the sum of first 10 terms.	04	OR	A geometric sequence has first term $a_1 = 3$ and common ratio $r = 2$: **a.** Find the 6th term. **b.** Find the sum of first 6 terms.	04
(xi)	Given $f(\theta) = cos2\theta$, **a.** Determine whether $f(\theta)$ is even, odd or neither **b.** Find the period of $f(\theta)$.	04	OR	Solve the following system of linear equations with complex coefficients. $z + (1+i)\,w = 2 + i;\ \ (1+i)\,z + w = 2 + i$	04
(xii)	Consider the letters of the word MANGO. **a.** Find number of ways to arrange all the letters. **b.** How many of these arrangements begin with M?	04	OR	Apply cross product to find angle between two vectors $\underline{a} = \hat{\imath}$ and $\underline{b} = \frac{1}{2}\hat{\imath} + \frac{\sqrt{3}}{2}\hat{\jmath}$	04

SECTION – C (Marks 32)

Note: Solve the following Questions. **(4 x 8 = 32)**
(Use of the graph paper is not necessary. Candidates can makes their own grid on answer book)

Q.3	The vertices of a tetrahedron are $A(1,2,3), B(2,31), C(3,1,2)$ and $D(4,2,1)$ **a.** Find volume of the tetrahedron $ABCD$ **b.** Using same vectors, find the volume of the parallelepiped formed by $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$.	08	OR	Prove the fundamental law of trigonometry $cos(\alpha - \beta) = cos\alpha\ cos\beta + sin\alpha\ sin\beta$ where α, β (real angles) are in standard position and $\alpha > \beta$.	08
Q.4	Solve the homogeneous system of equations using Gaussian Elimination method: $x + 2y - z = 0,\quad 2x + 3y - 2z = 0,$ $-x - y + z = 0$	08	OR	Given that $5, (5+3)2, (5+6)2^2, (5+9)2^3, \ldots$ **a.** Show that given sequence is arithmetic geometric. **b.** Find the sum of first n-terms of the sequence.	3+5
Q.5	Use binomial theorem to expand $\sqrt{101}$ and $\sqrt{99}$ separately. Multiply the two expansions and approximate the final answer rounded to three decimal places.	08	OR	A college has 8 Science teachers and 6 Arts teachers. In how many ways can a committee of 6 teachers be formed if it includes: **a.** Exactly 4 Science teachers and 2 Arts teachers **b.** 5 Science teachers and 1 Arts teacher **c.** At least 5 Science teachers **d.** At most 1 Arts teacher	2x4
Q.6	Draw the graph of $y = 2sin2x$; where $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$	08	OR	If x is very small such that its square and higher powers can be neglected, then show that: $\frac{(8+2x)^{\frac{2}{3}}}{(2+x)(\sqrt{4-2x})} \approx 1 - \frac{x}{12}$	6+2